# Swati Aggrawal

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## PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate from Sapienza University of Rome with a strong foundation in software engineering, cloud-based deployment, and applied AI/ML. Experienced in developing AI, computer vision, and retrieval-augmented generation (RAG) solutions using Python, PyTorch, TensorFlow, FastAPI, React, Docker, and AWS. Conducted research on Vision Transformers and continual learning, with an IEEE publication, and built and deployed an end-to-end RAG chatbot integrating LLM APIs and vector databases. Passionate about applying AI to solve real-world industrial challenges.

## SKILLS

- **Programming Languages:** Python, C++, Java, SQL, Bash
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, Generative AI, LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Vector Databases (Pinecone), Gemini API
- **Software Engineering:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), FastAPI, REST APIs, React, Git
- **Cloud & DevOps:** Docker, Cloud Deployment (AWS ECS, Fargate, CloudWatch, Render), Azure, Linux
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib

## EDUCATION

**Sapienza Università di Roma** **2026**

*Master of Science in Computer Science*

- Relevant Coursework: Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning.
- Developed and evaluated adapter-based continual learning methods for Vision Transformers using PyTorch and Elastic Weight Consolidation (EWC) to mitigate catastrophic forgetting.

**Sharda University** **2023**

*Bachelor of Technology – CSE with AI and ML*

- Relevant Coursework: Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- Thesis: Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded merit-based scholarship (Top 5–10% of cohort).

## EXPERIENCE

**Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis)** **Nov 2025 – May 2026**

https://github.com/swatiaggrawal/Thesis-Vision-Transformers

- Conducted research under Prof. Marco Raoul Marini on continual learning for computer vision in the Vision Lab.
- Developed and optimized PyTorch training and evaluation pipelines for continual learning experiments on Vision Transformers, benchmarking accuracy, forgetting, and computational efficiency across sequential classification tasks.
- Designed and implemented adapter-based learning and Elastic Weight Consolidation (EWC) techniques, achieving 90.59% average accuracy with only 2.01% average forgetting on Split CIFAR-100 (5 sequential tasks) — a ~97% relative reduction in catastrophic forgetting versus a CNN baseline.
- Benchmarked against ResNet-18 and naive ViT fine-tuning baselines (24.22% and 41.91% average accuracy respectively), validating results via cross-seed evaluation.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.
- Technologies: Python, PyTorch, Vision Transformers, Computer Vision

## PUBLICATION

**Machine Learning for Air Pollution — IEEE Conference Publication** **2023**
doi.org/10.1109/ViTECoN58111.2023.10157028

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

## PROJECTS

**Knowledge Copilot — Grounded RAG Chatbot**
github.com/swatiaggrawal/RAG-Chatbot | Live demo: rag-chatbot-new.onrender.com

- Built an end-to-end retrieval-augmented generation (RAG) chatbot that answers questions over FastAPI's official documentation, with every response grounded in retrieved source text and returned with per-answer citations to prevent hallucination.
- Architected a modular four-stage pipeline (ingestion, embedding, retrieval, generation), each independently runnable and testable via CLI, isolating failures to a specific stage — chunking, retrieval, or generation.
- Implemented recursive paragraph/sentence-aware chunking with Unicode normalization to fix PDF ligature artifacts, improving embedding quality over naive fixed-size character splitting.
- Integrated Gemini API embeddings (task-type-aware asymmetric embeddings for queries vs. documents) with Pinecone as a managed vector store, removing a local sentence-transformers/torch dependency to fit the deployed backend within a 512MB free-tier RAM limit.
- Built a FastAPI backend exposing a /chat endpoint and a build-free React (CDN) frontend displaying source citations per answer; implemented self-indexing startup logic that auto-runs the ingest→embed→store pipeline on a fresh deploy.
- Technologies: Python, FastAPI, React, Gemini API, Pinecone, RAG, LLMs, Vector Databases

**DocDelta — AI-Powered Document Intelligence & Comparison Platform**
github.com/swatiaggrawal/DocDelta

- Developed an end-to-end semantic document comparison API using FastAPI and Sentence Transformers (all-MiniLM-L6-v2), detecting and classifying sentence-level changes (added/removed/modified, numerical vs. wording) beyond traditional character-level diffs.
- Designed scalable embedding and cosine-similarity retrieval pipelines, achieving ~840ms warm-request latency on CPU-only inference.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring for latency and error tracking.
- Built REST APIs with async request handling and validation, optimizing container size (CPU-only PyTorch builds) and eliminating cold-start overhead via model preloading.
- Technologies: Python, FastAPI, Sentence Transformers, PyTorch, Docker, AWS ECS Fargate, CloudWatch, REST APIs

**Metal Defect Detection using Computer Vision**
github.com/swatiaggrawal/Metal-Defect-Detection

- Built a custom CNN (TensorFlow/Keras) plus SVM and Random Forest classifiers (HOG features) for automated industrial metal surface defect detection across 6 defect classes on the NEU dataset.
- Built image preprocessing, augmentation, training, and evaluation pipelines, tuning classical models via GridSearchCV.
- Achieved 97.22% test accuracy with the CNN versus 86.11% for both SVM and Random Forest — an ~11-point improvement demonstrating the value of learned spatial features over hand-crafted descriptors.
- Evaluated models using accuracy, confusion matrices, precision, recall, and F1-score.
- Technologies: Python, TensorFlow, Keras, OpenCV, Scikit-Learn